Parampreet Singh

Data Engineer II
(864) 207-1256 ◆ paramps@g.clemson.edu ◆ LinkedIn ◆ Github

Professional Summary

Versatile Data professional with over 4 years of experience in designing and optimizing solutions for large-scale data. Strong analytical skills, capable of deriving insights from complex datasets to support business decision-making. Proven leadership in managing teams and driving projects to successful completion with a proactive approach.

Professional Experience

Clemson University — **Clemson, SC**

Data Engineer — 01/2024 – 08/2024

- **Boosted predictive accuracy** of vacancy enthalpies by 15% through a modified CGCNN, leveraging residual blocks, custom convolutional layers and transfer learning, resulting in an R2 score of over 0.89.
- **Accelerated DFT-relaxed crystal structure prediction** and cut computational costs by 7x by developing a deep learning model using Graph Convolutional Networks and Generative Adversarial Networks.
- **Designed and optimized a data pipeline** in PyTorch, enabling efficient handling of variable-sized graph data.

IT Data Engineer — 11/2022 – 12/2023

- **Achieved a 25% reduction** in unexpected IT asset downtime through collaborative development and deployment of an Azure-based predictive maintenance model.
- **Engineered an automated data pipeline in Azure Data Factory**, integrating weekly preprocessing and batch inferencing to deliver actionable insights for proactive asset maintenance, connected to a Power BI dashboard.
- Accelerated IT ticket resolution, leveraging **Azure OpenAI** integration for an IT assistant chatbot development.
- Streamlined data preprocessing using **Azure ETL and NLP**, improving IT support efficiency for 5,000+ users.

Maruti Suzuki India Ltd — **Gurugram, IND**

Data Engineer II — 10/2021 – 07/2022

Collaborated with sales and marketing to transform PAN-India sales data, supporting predictive lead prioritization.

- **Collaborated on developing a scalable data pipeline using Azure ML**, enhancing lead prioritization and vehicle inventory planning, processing over 10,000 sales opportunities daily.
- **Improved prediction accuracy by 4%** monthly through bi-weekly retraining and A/B testing in Azure ML, ensuring adaptability to evolving automotive sales patterns.
- **Migrated legacy systems (Hadoop, Hive) to a cloud ecosystem**, optimizing data acquisition workflows into ADLS gen2, achieving 99.8% uptime and reducing processing times by 45%.
- **Enhanced data governance and compliance** using Unity Catalog in Databricks while executing high-performance transformations with Apache Spark for sales data analytics.
- **Managed SDLC with agile workflows**, UAT, and hypercare, mentoring team members on advanced data engineering practices and fostering a collaborative environment.

Data Engineer I — 07/2020 – 10/2021

Served as a subject matter expert in automotive sales telemetry data quality and predictive analytics.

- **Developed and optimized data pipelines** to structure large volumes of sales telemetry data into Delta tables in Synapse, ensuring data consistency and integrity for downstream use.
- **Delivered a cost-efficient solution saving $1.2M** by improving data ingestion pipelines and repository frameworks tailored for automotive sales analytics.
- **Built scalable pipelines to deploy and monitor ML models** in production environments, enabling daily batch predictions for sales forecasting.

- Established data-driven development practices by **operationalizing key performance metrics**, improving decision-making for vehicle inventory and sales teams.
- Proactively debugged data issues, identified root causes, and **implemented robust solutions** to ensure reliability in automotive sales forecasting pipelines.

Mahindra Research Valley — **Chennai, IND**
Data Analyst Intern — 01/2019 – 05/2019

- **Attained a 92% R2 score**, forecasting axle power loss split, using predictive analytics & mathematics in MS Excel.
- Proposed design improvements, reducing axle power loss by 3%, published the findings in SAE International. [>]

PROJECTS

- **ResumeRAG:** Developed an AI-powered resume generator using LangChain, Chroma, and RAG, leveraging hybrid search to enable fast, efficient, and personalized query-based resume interactions.
- **LSTM Attention:** Achieved a 60% reduction in SoftMax loss by implementing an LSTM with Attention model, significantly outperforming standard models without attention.
- **Predicting Hearing Thresholds from Brain MRI Scans:** Leveraged CNNs, SVR, PCA, and XGBoost to predict hearing thresholds from brain MRI scans, achieving a MAE of 1.14 and MSE of 21.04.

SKILLS

Machine Learning, Deep Learning, Data Modelling & Visualisation, ETL/ELT, Ad Hoc Data Analysis, Statistics

Programming: SQL, Python, R, Scala, Matlab, Big Query
Databases: MySQL, Oracle, SQL Server, MongoDB
Cloud Platforms: Azure, GCP, Databricks, AWS
ETL/ ELT: DataLake, Warehouse, CI/CD, Snowflake
Big Data: Spark, Kafka, Hadoop, Airflow, DBT, NLP, GenAI
BI Tools: Tableau, Power BI, Ggplot, Snowflake

CERTIFICATIONS

AWS Machine Learning Specialty
PowerBI Desktop for Business Intelligence

EDUCATION

Clemson University — **South Carolina, SC**
Master of Science, Computer Science — 08/2024

Punjab Engineering College — **Chandigarh, IND**
Bachelor of Technology, Mechanical Engineering — 06/2020

AWARDS & ASSOCIATIONS

- Graduate Research Assistantship, Clemson University